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Letter to Glyco-forum

GlyGen: Computational and Informatics Resources for Glycoscience

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Methods to obtain data relevant to glycobiology are rapidly evolving, thereby providing enormous opportunities to significantly increase knowledge, insight, and understanding in this important domain. Nevertheless, progress in this field is compromised by the lack of glycoinformatics databases and tools that combine information from related disciplines, including genetics, proteomics, pathology, etc. GlyGen is an international initiative funded by the National Institutes of Health Common Fund (Contract #1U01GM125267–01) and aimed at developing an integrated, extendable, and cross-disciplinary resource, which provides tools and data to address glycoscience questions that can currently be answered only by extensive literature-based research and manual browsing and data collection from diverse resources. Here, we announce the release of the GlyGen Portal (https://www.glygen.org), which provides a user-friendly interface that facilitates exploration of glycoscience data in the context of other information that we have compiled, harmonized, and integrated from diverse international bioinformatics resources, including the National Center for Biotechnology Information (NCBI), UniProt, the Protein Data Bank (PDB), UniCarbKB, and the GlyTouCan glycan structure repository.

Development of the infrastructure required for the compilation, integration, and display of these diverse data is an immense challenge. We have now made sufficient progress to enable the first fully functional version of the GlyGen Portal (https://www.glygen.org),

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which is currently limited in its scope to include data relevant to *N*- and *O*-glycans from three species (human, mouse, and rat). We invite members of the biomedical community to explore and evaluate this site in the context of their own research and suggest enhancements, additional data to include, and more intuitive ways of browsing and visualizing the information we have integrated. All comments can be sent from the GlyGen web portal by navigating to "Help -> Contact Us" or via email to will@ccrc.uga.edu or mazumder@gwu.edu.

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